

CHEUNG CHAU, Hong Kong

WMO: 450440

Lat: 22.2011N

Lon: 114.0267E

Elev: 79

StdP: 100.38

Time Zone: 8.00 (E08)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.0	9.5	-1.4	3.4	11.8	1.4	4.2	12.6	12.2	10.1	11.3	10.1	8.6	0	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.5	32.7	28.2	32.0	27.9	31.3	27.6	29.1	31.4	28.6	30.9	28.1	30.4	4.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.4	25.0	30.6	27.9	24.3	30.1	27.5	23.7	29.6	95.9	31.6	93.3	31.1	91.3	30.6	33.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.7	10.8	9.5	DB	6.5	34.8	2.6	0.9	4.6	35.5	3.1	36.0	1.6	36.5	-0.3	37.2	
			WB	2.9	30.4	2.3	1.3	1.3	31.3	-0.1	32.1	-1.3	32.8	-3.0	33.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.3	16.1	17.2	19.1	22.7	25.9	27.7	28.3	28.1	27.7	25.6	22.2	18.0
	DBStd	5.03	2.85	3.66	2.93	2.62	2.09	1.61	1.39	1.48	1.58	1.79	2.66	2.96
	HDD10.0	4	2	2	0	0	0	0	0	0	0	0	0	1
	HDD18.3	209	77	57	26	3	0	0	0	0	0	0	4	42
	CDD10.0	4841	192	203	283	380	494	532	568	561	530	484	367	247
	CDD18.3	2004	9	25	51	133	236	282	309	303	280	226	121	31
	CDH23.3	16486	4	22	61	489	1805	2874	3377	3209	2742	1492	381	28
(13)	CDH26.7	4426	0	0	2	67	392	817	1091	981	761	281	34	1
(14)	Wind	WSAvg	4.9	4.9	4.6	4.6	4.5	4.8	5.1	4.9	4.6	5.2	5.4	5.4
(15) Precipitation	PrecAvg	2009	21	37	62	148	289	368	284	344	244	128	28	23
	PrecMax	3010	64	222	396	356	742	1128	712	792	662	783	117	154
	PrecMin	883	0	0	0	2	6	75	63	91	33	0	0	0
	PrecStd	539	18	61	81	105	184	245	149	200	170	184	36	41
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	25.4	26.6	30.0	32.8	32.8	33.2	33.7	33.6	31.9	29.0	25.6
		MCWB	19.2	21.3	23.4	26.1	27.9	28.7	28.8	28.0	27.8	26.0	24.0	20.9
	2%	DB	22.0	23.9	25.0	28.4	31.1	31.9	32.2	32.4	32.2	30.2	27.6	24.0
		MCWB	18.0	20.4	21.8	25.0	27.8	28.2	28.4	28.1	27.3	24.9	23.0	19.4
	5%	DB	20.7	22.4	23.7	27.2	30.0	31.1	31.6	31.6	31.3	29.1	26.5	22.7
		MCWB	17.2	19.7	21.0	24.2	27.0	28.0	28.1	27.9	26.7	24.2	22.2	18.6
	10%	DB	19.6	21.2	22.6	25.9	28.8	30.1	30.8	30.7	30.2	28.2	25.4	21.6
		MCWB	16.5	19.1	20.5	23.5	26.3	27.5	27.7	27.6	26.2	23.7	21.7	18.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.3	22.4	24.0	26.7	29.2	29.7	29.7	29.4	29.2	27.2	25.1	22.3
		MCDB	22.4	24.1	25.9	29.1	30.8	31.6	32.0	31.9	32.0	30.4	27.9	24.5
	2%	WB	19.1	21.4	22.7	25.6	28.2	28.8	29.0	28.7	28.1	26.2	24.2	21.1
		MCDB	20.7	22.8	23.9	27.4	30.3	30.8	31.4	31.2	31.1	28.9	26.3	22.7
	5%	WB	18.1	20.5	21.8	24.8	27.5	28.2	28.4	28.2	27.5	25.5	23.4	20.0
		MCDB	19.9	21.9	23.0	26.3	29.3	30.2	30.7	30.7	30.2	27.9	25.4	22.0
	10%	WB	17.2	19.4	21.1	24.2	26.8	27.8	28.0	27.7	26.9	24.8	22.6	18.8
		MCDB	19.1	21.0	22.3	25.6	28.3	29.6	30.2	30.0	29.2	27.1	24.7	21.1
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	4.8	4.6	4.5	4.5	4.2	4.5	4.6	4.9	5.1	5.2	5.6
		MCDBR	6.2	6.2	5.8	6.1	5.9	5.4	5.5	5.9	6.5	6.1	5.9	6.3
	5% WB	MCWBR	3.6	3.6	3.4	3.2	3.2	2.9	3.0	3.1	3.3	3.1	3.1	3.8
		MCWBR	5.1	5.6	5.2	5.6	5.3	4.9	5.2	5.5	5.9	5.1	5.1	5.4
(40) Clear-Sky Solar Irradiance	taub	taub	0.578	0.626	0.748	0.723	0.562	0.531	0.519	0.584	0.619	0.671	0.573	0.568
		taud	1.817	1.749	1.546	1.634	2.036	2.143	2.171	1.964	1.855	1.697	1.893	1.861
	Ebn at Noon	Ebn at Noon	683	682	622	651	760	778	788	740	705	638	683	672
		Edh at Noon	195	222	282	262	174	155	151	186	204	230	179	181
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.99	3.10	3.40	3.86	4.60	4.72	5.43	5.05	4.74	4.37	3.50	3.11
	RadStd	RadStd	0.50	0.50	0.45	0.56	0.57	0.68	0.57	0.45	0.39	0.43	0.40	0.46

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	+0.25	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page